

Q1: K-Means Clustering

Given a 4-digit student ID **ABCD** (each digit 0–9), define points:

- $P_1 = (A, B), P_2 = (C, D)$
- $P_3 = (A + C, B + D), P_4 = (|A - C|, |B - D|)$

Perform **K-means clustering** ($K = 2$) with initial centroids:

$$C_1 = P_1, C_2 = P_2$$

Tasks:

1. Compute distances using:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

2. Assign each point to the nearest centroid
3. Update centroids after one iteration

Q2: Neural Network (Feedforward + Backpropagation)

Given student ID **ABCD**:

- Inputs: $x_1 = A, x_2 = B$
- Weights:
 - Input→Hidden: $w_1 = 0.A, w_2 = 0.B, w_3 = 0.C, w_4 = 0.D$
 - Hidden→Output: $w_5 = 0.(A + B), w_6 = 0.(C + D)$
- Biases: Hidden = 0.1, Output = 0.2
- Target: $T = (A + B + C + D) \bmod 2$

Activation:

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Tasks:

1. Perform feedforward to compute output
2. Compute error: $E = \frac{1}{2}(T - O)^2$
3. Perform one backprop step and update w_5, w_6 with $\eta = 0.1$